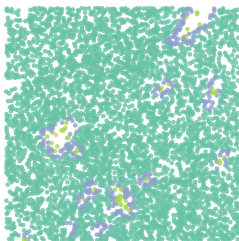
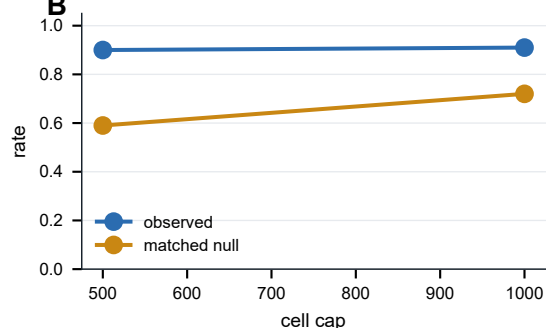


Figure 5 | SNRA rescues defensible claims and rejects block-sensitive claims

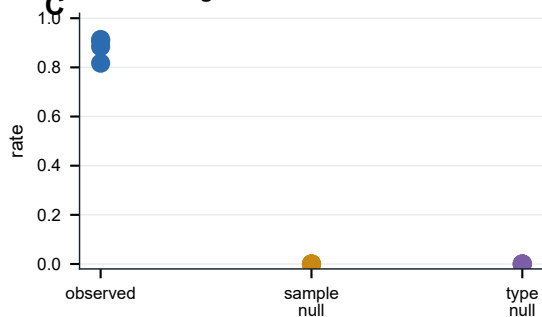
A Tissue map / spatial context



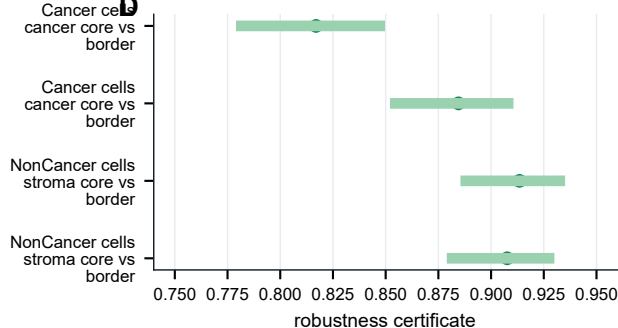
B Uncalibrated niche claim



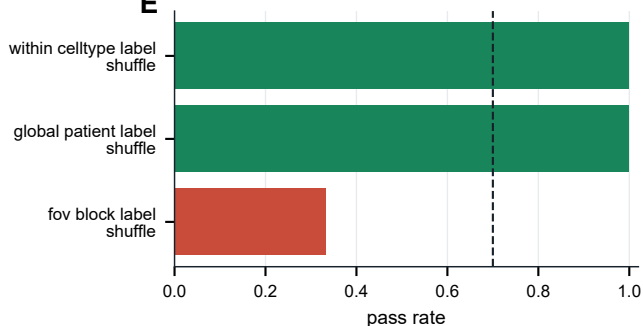
C Matched negative-control distribution



D Robustness score



E Surviving niche or rejected claim



F Corrected biological interpretation

Before SNRA: broad tumor compartment niche

After SNRA: lineage-stratified boundary robustness

Blocked: cell-level morphology validation from BCC